PRESENTATION ON SOFTWARE DEFINED NETWORK (SDN)

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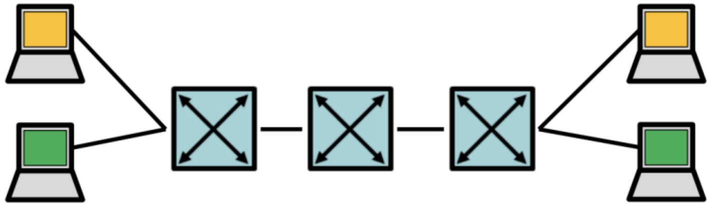
INTRODUCTION (PROJECT-3)

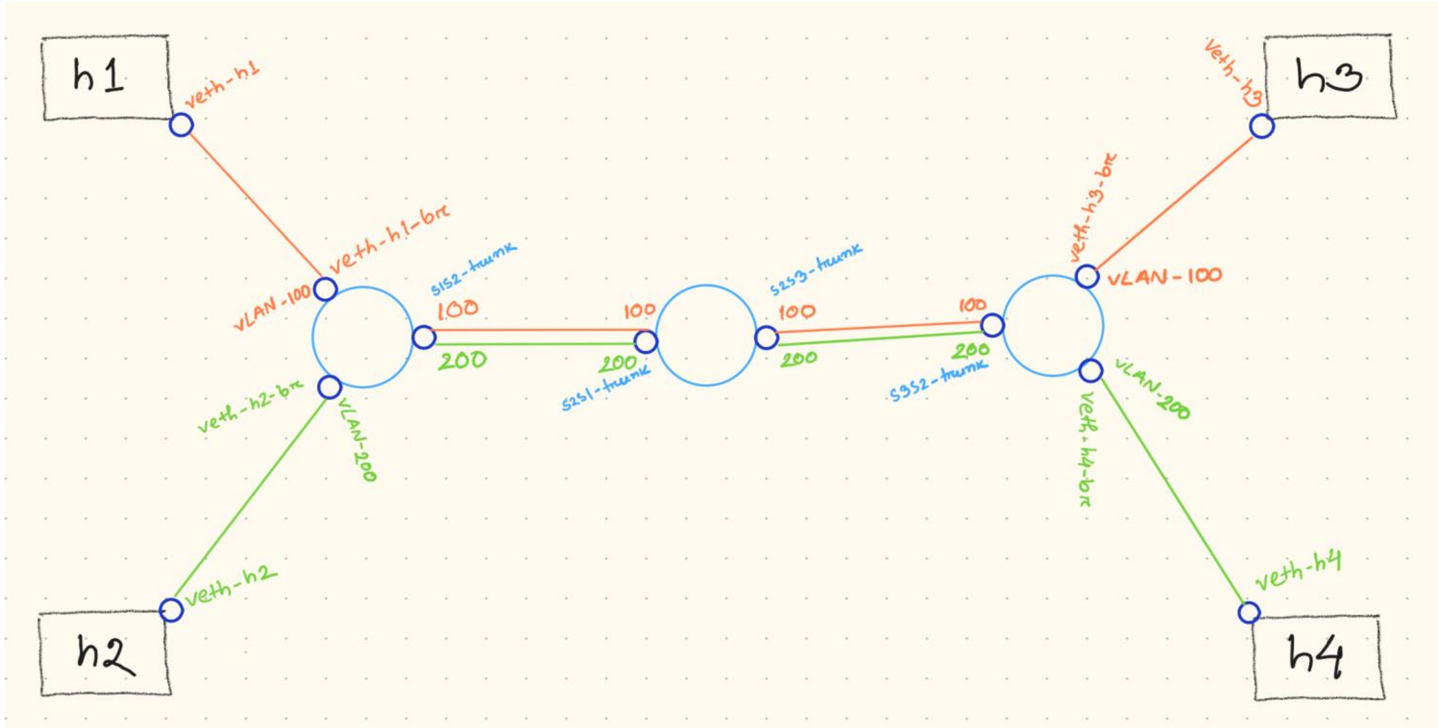
- Implement with namespaces and Open vSwitch
  - Two VLANs
  - Enable ping only among hosts in same VLAN



- Using Mininet
  - Implement a similar network with an arbitrary number of switches
- Report ping and iperf measurements
- Check a topology with 5 switches on the ONOS web-GUI

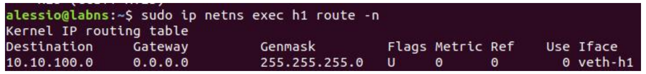
TERMINOLOGIES

- Linux's native network virtualization features
- Mininet as network emulator
- SDN
- ONOS as SDN Controller



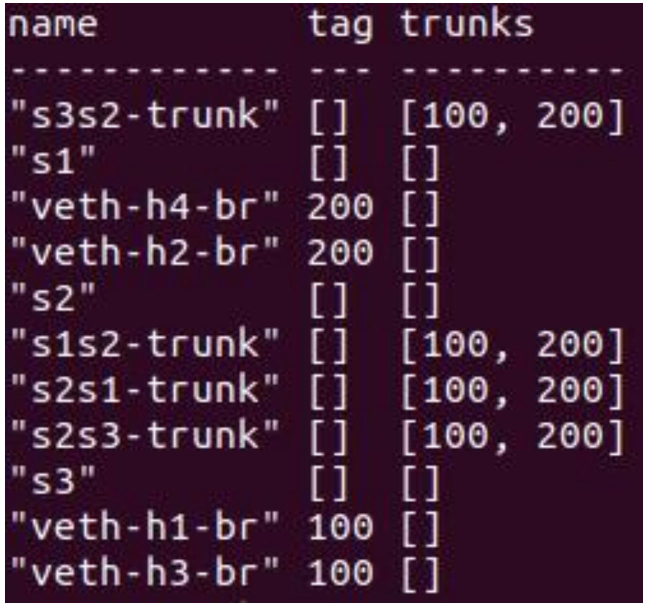
FIRST PART OF THE PROJECT

- Used Linux’s native network virtualization features to create namespaces and open vSwitches
- Only Layer-2 switches in the network, no router
- Hosts are separated by two VLANs(100,200)
- No connectivity to internet from ns
- No route from host to ns



```
alessio@labns:~$ sudo ip netns exec h1 route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
10.10.100.0     0.0.0.0         255.255.255.0   U     0      0        0 veth-h1
```

```
alessio@labns:~$ sudo ip netns exec h1 arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
10.10.100.3              ether   2e:9c:a7:56:95:08   C                     veth-h1
```



```
name           tag trunks
-------------- --- ----------
"s3s2-trunk"   []  [100, 200]
"s1"           []  []
"veth-h4-br"   200 []
"veth-h2-br"   200 []
"s2"           []  []
"s1s2-trunk"   []  [100, 200]
"s2s1-trunk"   []  [100, 200]
"s2s3-trunk"   []  [100, 200]
"s3"           []  []
"veth-h1-br"   100 []
"veth-h3-br"   100 []
```

SECOND PART OF THE PROJECT (MININET)

- Emulate the same network with arbitrary number of switches
- Running the topology with mininet
- Test desired host connectivity (test pings)
- Behaves same from the topology earlier

```
alessio@labns:~/NETM-scripts-git/project-3$ sudo python3 mininet-topo.py
*** Creating network
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1 s2 s3
*** Adding links:
(h1, s1) (h2, s1) (h3, s3) (h4, s3) (s1, s2) (s2, s3)
*** Configuring hosts
h1 (cfs -1/100000us) h2 (cfs -1/100000us) h3 (cfs -1/100000us) h4 (cfs -1/100000us)
*** Starting controller
onos
*** Starting 3 switches
s1 s2 s3 ...
*** Applying VLAN tagging and trunks
*** Setup complete; dropping to CLI
*** Starting CLI:
mininet>
mininet>
mininet> h1 ping h2
connect: Network is unreachable
mininet> h1 ping h3
PING 10.10.100.3 (10.10.100.3) 56(84) bytes of data.
64 bytes from 10.10.100.3: icmp_seq=1 ttl=64 time=82.6 ms
64 bytes from 10.10.100.3: icmp_seq=2 ttl=64 time=24.5 ms
^C
--- 10.10.100.3 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1006ms
rtt min/avg/max/mdev = 24.563/53.598/82.634/29.036 ms
mininet>
mininet> h2 ping h4
PING 10.10.200.4 (10.10.200.4) 56(84) bytes of data.
64 bytes from 10.10.200.4: icmp_seq=1 ttl=64 time=41.3 ms
64 bytes from 10.10.200.4: icmp_seq=2 ttl=64 time=39.8 ms
```

CHECKING THE ROUTE

- Check the route from h1 and h3 (how many hops and why)
- Check the ARP resolution of h1 or h3
- Check the routing table(cannot ping internet)

```
mininet> h1 traceroute h3
traceroute to 10.10.100.3 (10.10.100.3), 30 hops max, 60 byte packets
 1  10.10.100.3 (10.10.100.3)  128.416 ms  136.373 ms  149.645 ms
mininet>
```

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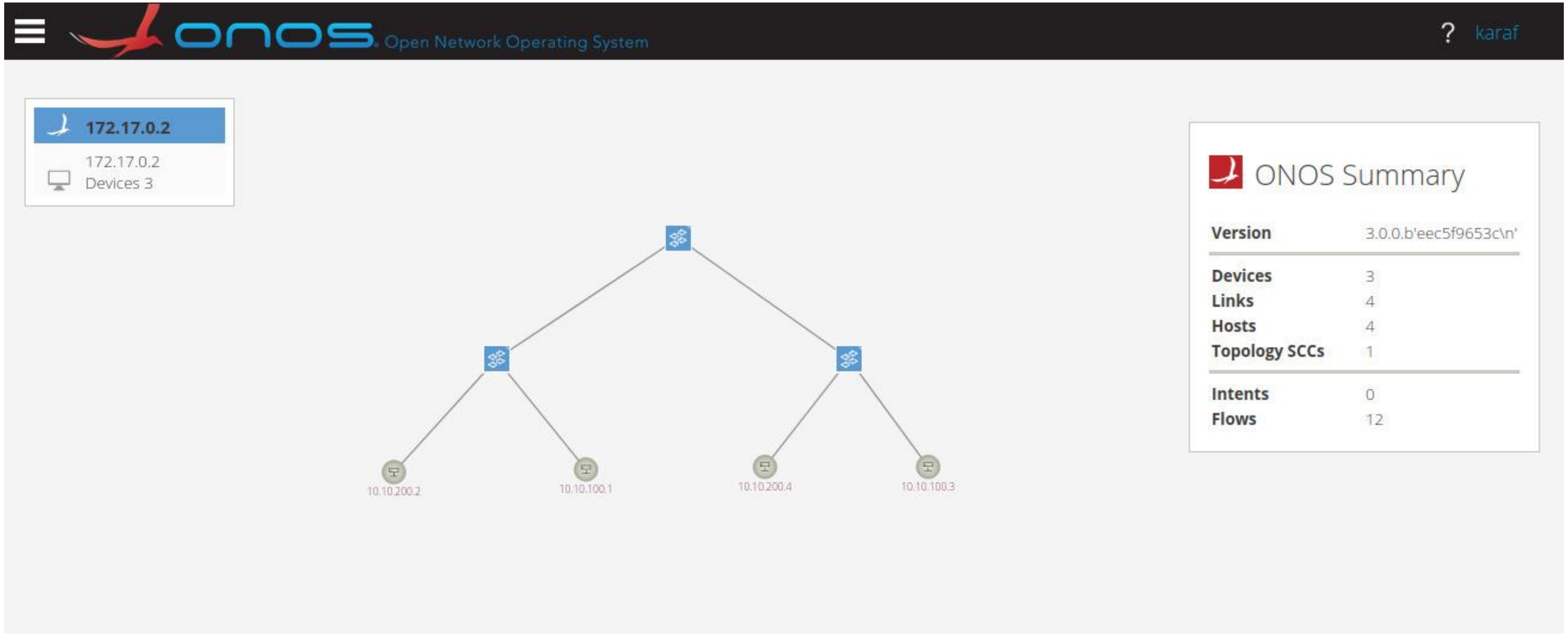
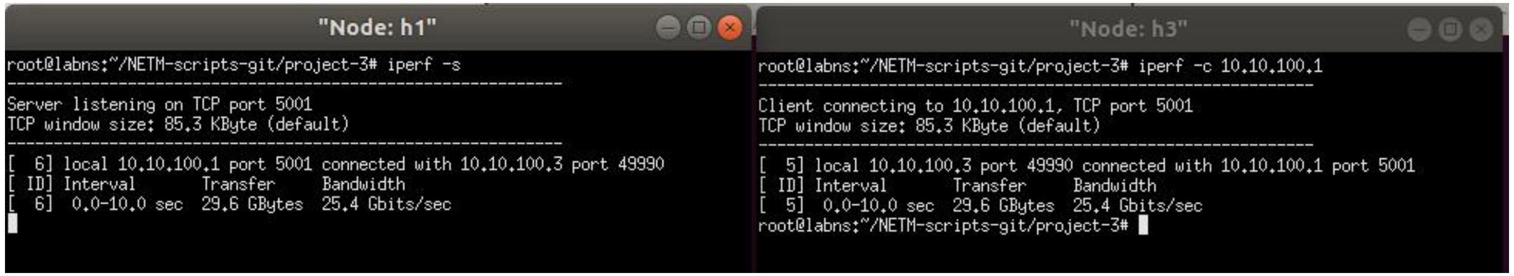
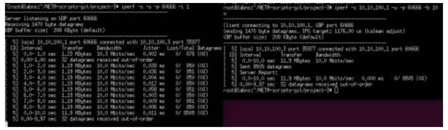
INSPECTING TOPOLOGY ON ONOS

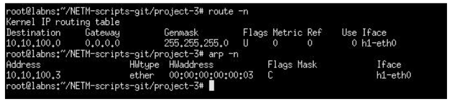


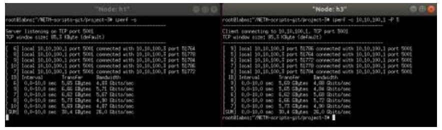
MEASURING BANDWIDTH BETWEEN HOSTS(IPERF)

- Iperf measurements between h1 as a server and h3 as a client
- Testing with UDP
- Testing with set bandwidth
- Testing with multiple sessions









THANK YOU